$$P(E_2/A) = \frac{P(E_2)P(A/E_2)}{\sum_{i=1}^{2} P(E_i)P(A/E_i)}$$

$$= \frac{P(E_2)P(A/E_2)}{P(E_1)P(A/E_1)+P(E_2)P(A/E_2)}$$

$$= \frac{\frac{1}{2}\cdot\frac{5}{11}}{\left(\frac{1}{2}\times\frac{3}{7}\right)+\left(\frac{1}{2}\times\frac{5}{11}\right)} = \frac{35}{68}$$

Example 8.32

X speaks truth 4 out of 5 times. A die is thrown. He reports that there is a six. What is the chance that actually there was a six?

Solution:

Let us define the following events.

E_1 : X speaks truth

E_2 : X tells a lie

E : X reports a six

From the data given in the problem, we have

$$P(E_1) = \frac{4}{5}; P(E_2) = \frac{1}{5}$$

$$P(E/E_1) = \frac{1}{6}; P(E/E_2) = \frac{5}{6}$$

The required probability that actually there was six (by Bayes theorem) is

$$P(E_1/E) = \frac{P(E_1)P(E/E_1)}{P(E_1)P(E/E_1)+P(E_2)P(E/E_2)}$$

$$= \frac{\frac{4}{5}\times\frac{1}{6}}{\left(\frac{4}{5}\times\frac{1}{6}\right)+\left(\frac{1}{5}\times\frac{5}{6}\right)} = \frac{4}{9}$$

Example 8.33

A factory has 3 machines A_1, A_2, A_3 producing 1000, 2000, 3000 screws per day respectively. A_1 produces 1% defectives, A_2 produces 1.5% and A_3 produces 2% defectives. A screw is chosen at random at the end of a day and found defective. What is the probability that it comes from machines A_1?

Solution:

$P(A_1)$ = P(that the machine A_1 produces screws) = $\frac{1000}{6000} = \frac{1}{6}$

$P(A_2)$ = P(that the machine A_2 produces screws) = $\frac{2000}{6000} = \frac{1}{3}$

$P(A_3)$ = P(that the machine A_3 produces screws) = $\frac{3000}{6000} = \frac{1}{2}$

Let B be the event that the chosen screw is defective

$\therefore P(B/A_1)$ = P(that defective screw from the machine A_1) = 0.01

$P(B/A_2)$ = P(that defective screw from the machine A_2) = 0.015 and

$P(B/A_3)$ = P(that defective screw from the machine A_3) = 0.02

We have to find $P(A_1/B)$

Hence by Baye's theorem, we get

$$P(A_1/B) = \frac{P(A_1)P(B/A_1)}{P(A_1)P(B/A_1)+P(A_2)P(B/A_2)+P(A_3)P(B/A_3)}$$

$$= \frac{\left(\frac{1}{6}\right)(0.01)}{\left(\frac{1}{6}\right)(0.01)+\left(\frac{1}{3}\right)(0.015)+\left(\frac{1}{2}\right)(0.02)}$$

$$= \frac{0.01}{0.01+0.03+0.06} = \frac{0.01}{0.1} = \frac{1}{10}$$

Exercise 8.2

1. A family has two children. What is the probability that both the children are girls given that at least one of them is a girl?

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